

SQL

SQL Introduction

Standard language for querying and manipulating data

Structured Query Language

Many standards out there:

- ANSI SQL, SQL92 (a.k.a. SQL2), SQL99 (a.k.a. SQL3),

SQL

- Data Definition Language (DDL)
 - Create/alter/delete tables and their attributes
 - Following lectures...
- Data Manipulation Language (DML)
 - Query one or more tables – discussed next !
 - Insert/delete/modify tuples in tables

Table name

Attribute names

Tables in SQL

Product

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

Tuples or rows

Tables Explained

- The *schema* of a table is the table name and its attributes:

Product(PName, Price, Category, Manufacturer)

- A *key* is an attribute whose values are unique; we underline a key

Product(PName, Price, Category, Manufacturer)

Data Types in SQL

- Atomic types:
 - Characters: CHAR(20), VARCHAR(50)
 - Numbers: INT, BIGINT, SMALLINT, FLOAT
 - Others: MONEY, DATETIME, ...
- Every attribute must have an atomic type
 - Hence tables are flat
 - Why ?

Tables Explained

- A tuple = a record
 - Restriction: all attributes are of atomic type
- A table = a set of tuples
 - Like a list...
 - ...but it is unordered:
no **first()**, no **next()**, no **last()**.

SQL Query

Basic form: (plus many many more bells and whistles)

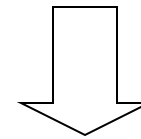
```
SELECT <attributes>  
FROM   <one or more relations>  
WHERE  <conditions>
```


Simple SQL Query

Product

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

```
SELECT *  
FROM Product  
WHERE category='Gadgets'
```



“selection”

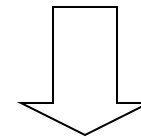
PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks

Simple SQL Query

Product

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

```
SELECT PName, Price, Manufacturer
FROM Product
WHERE Price > 100
```



“selection” and
“projection”

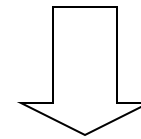
PName	Price	Manufacturer
SingleTouch	\$149.99	Canon
MultiTouch	\$203.99	Hitachi

Notation

Input Schema

Product(PName, Price, Category, Manufacturer)

```
SELECT PName, Price, Manufacturer
FROM   Product
WHERE  Price > 100
```



Answer(PName, Price, Manufacturer)

Output Schema

Details

- Case insensitive:
 - Same: SELECT Select select
 - Same: Product product
 - Different: 'Seattle' 'seattle'
- Constants:
 - 'abc' - yes
 - "abc" - no

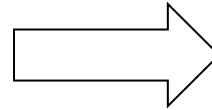
The **LIKE** operator

```
SELECT *  
FROM Products  
WHERE PName LIKE '%gizmo%'
```

- s **LIKE** p: pattern matching on strings
- p may contain two special symbols:
 - % = any sequence of characters
 - _ = any single character

Eliminating Duplicates

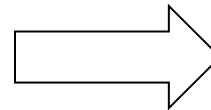
```
SELECT DISTINCT category  
FROM Product
```



Category
Gadgets
Photography
Household

Compare to:

```
SELECT category  
FROM Product
```



Category
Gadgets
Gadgets
Photography
Household

Ordering the Results

```
SELECT pname, price, manufacturer  
FROM Product  
WHERE category='gadgets' AND price > 50  
ORDER BY price, pname
```

Ties are broken by the second attribute on the ORDER BY list, etc.

Ordering is ascending, unless you specify the DESC keyword.